

Debdip Bandyopadhyay

Senior SDET | Lead AI Quality Engineer (Designation: Developer / Project Associate)

Kolkata, India | +91-8285578544 | debdip1992@outlook.com | Portfolio: [debdiparvr.github.io/Fun_projects/portfolio](https://github.com/debdiparvr/Fun_projects/portfolio) | LinkedIn | GitHub
Interactive Technical Artifacts: HF Telemetry Hub (cloze-congruence-ai-forensics-hub-v1) | ScribeMark Text (scribemark.streamlit.app) | ScribeMark Vision (scribemarkimage.streamlit.app)

PROFESSIONAL SUMMARY

Hybrid Senior SDET and AI Evaluation Engineer with **11+ years** of production engineering experience connecting enterprise CI/CD reliability with mathematical AI research across Generative AI platforms, safety-critical Aerospace systems, and global Oracle ERP ecosystems. Architect of automated CI evaluation gates in **Azure DevOps (DeepEval, LLM-as-a-Judge, Playwright, Pytest-BDD, pytest-xdist)** enforcing deterministic release thresholds across non-deterministic LLM and RAG applications. Holds an **M.Tech from IIT Jodhpur (School of AI & Data Science)** with first-author practical IP in *Cloze Congruence (100.00% Top-1 LLM attribution; 98.92% 10-language AUROC)* and *Latent Resonance Diffusion Forensics (100.0% AUROC on N=1,000 images)*, directly mitigating enterprise AI vulnerabilities including training data leakage, shadow model attribution, synthetic media injection, and hallucination bounds (CERN Zenodo DOIs: [10.5281/zenodo.22158286](https://zenodo.org/record/22158286) & [10.5281/zenodo.22158483](https://zenodo.org/record/22158483)).

TECHNICAL SKILLS (GROUPED BY ARCHITECTURAL FUNCTION)

- AI Quality & LLM Evaluation:** DeepEval, RAG Triad (Faithfulness, Answer Relevancy, Context Recall), LLM-as-a-Judge (Azure OpenAI), Hallucination Detection, Semantic Similarity, Non-Deterministic Output Validation, Foundation Model Provenance & Multimodal Forensics, NVIDIA NIM, PyTorch.
- Automation Engineering:** Python, Playwright, Pytest, Pytest-BDD, Page Object Model (POM), Parallel Execution (pytest-xdist), Selective Test Sharding, Cypress, REST API Automation (Requests, Postman v12), Allure Interactive Reporting.
- CI/CD & LLMops Platform:** Azure DevOps Pipelines, Automated Evaluation Gates, Docker, Poetry, Git, Linux, Unattended Nightly Execution & Runtime State Persistence.
- Enterprise & Legacy Ecosystems:** Oracle EBS (AP, AR, PO, CE, FA), Tricentis Tosca, UFT, SAP MM, Jira Governance (JQL Coverage & Defect Telemetry).

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Apr 2021 – Present

Official HR Record — Designation: *Developer / Project Associate* (Employee ID: 932565 | DOJ: 19 April 2021)

Kolkata, India

Role 1: Senior SDET / Lead AI Quality Engineer — GenAI CoE (Designation: Developer / Project Associate)

Sep 2025 – Present

- GenAI CI/CD Evaluation Gates:** Architected automated CI evaluation gates in **Azure DevOps** using **DeepEval** and **Azure OpenAI (LLM-as-a-Judge)**, blocking builds on hallucination and context-drift thresholds across **15+ conversational flows** while enforcing strict JSON schema criteria across the RAG Triad (Faithfulness ≥ 0.85 , Answer Relevancy ≥ 0.90 , Context Recall) and Semantic Similarity.
- Framework Architecture:** Engineered an enterprise-grade **Playwright + Python + Pytest-BDD** automation framework adopting modular Page Object Model (POM) abstractions and **Requests** REST API contract validators for full-stack generative AI applications.
- Speed & Infrastructure Leverage:** Achieved a **40% reduction** in regression cycle times through **parallel execution with pytest-xdist, test sharding, and optimized headless execution** inside containerized **Docker** environments with Allure interactive telemetry.
- Cross-Domain Engineering Leadership:** Mentored **10+ automation engineers** on Pythonic evaluation patterns and partnered with NLP Scientists, Cloud Architects, and Product Owners to establish synthetic evaluation datasets and zero-trust AI quality benchmarks (conferred **Cognizant Asset Creation & Pride Awards**).

Role 2: Senior Automation Engineer / Quality Consultant — Oracle Enterprise Systems (Designation: Developer / Sr. QEA)

Apr 2021 – Aug 2025

- Multi-OU Fault-Tolerant ERP Architecture:** Designed and deployed a multi-module, multi-Operating Unit (OU) automation framework across **Oracle EBS (AP, AR, PO, CE, FA)** using **UFT, Tricentis Tosca, and Python**, implementing persistent runtime state stores that enabled **automated resume-from-failure** and unattended nightly runs across **APAC, EMEA, LATAM, and North America** deliveries.
- Global CPU Patch Cycle Execution (Wyndham Destinations):** Migrated **250 legacy financial test cases (183 AP, 67 PO)** and executed **591 automated regression scripts (285 AP, 52 CE, 72 FA, 94 PO & RI, 88 AR)** across quarterly NTST/NUAT CPU patch cycles with minimal SME dependency.
- Enterprise Client PoCs & Data Engineering (American Express):** Delivered an input-driven **Tricentis Tosca PoC (PO/PR workflows)** for **American Express** closed per schedule; built automated user/approver hierarchy extraction, **.xls-to-.xlsx** migration pipelines, and dynamic 180-day period-based test data randomization with error-logging runner sheets.
- Jira Telemetry & Governance:** Defined Jira dashboard rules, defect labeling standards, and custom **JQL coverage queries** for release tracking; delivered recorded cross-team KTs on multi-module framework debugging patterns.

The Boeing Company

Oct 2018 – Apr 2021

Software Engineer – Test Automation

Bengaluru, India

- Safety-Critical Aerospace Verification:** Engineered safety-critical test suites across **2,000+ test cases (Boeing Pride Award)** for regulated avionics and flight-operations software, leading a 3-engineer automation cell and maintaining strict requirement-to-test traceability across multi-phase release cycles.
- Regression Architecture & Code Governance:** Refactored enterprise regression libraries to eliminate non-deterministic flakiness, integrated CI pipeline execution, conducted peer code reviews, and mentored junior engineers across Systems Engineering and QA squads.
- Mission-Critical Recognition:** Awarded both the **Boeing Pride Award (737 MAX return-to-service validation)** and **Boeing Star Award (2019)** for engineering rigor and high-velocity validation across compressed sprint schedules.

PROFESSIONAL EXPERIENCE (CONTINUED)

Accenture LLP Test Automation Engineer	Oct 2014 – Oct 2018 Kolkata / Noida, India
• Licensing Cost Reduction Utility: Reduced third-party licensing expenses by ~\$8,000 annually via 'One Click Capture' (CIO India featured) and cut defect evidence packaging time by 75% by engineering an in-house multi-screen state-capture and document compilation utility. • Multi-Platform Enterprise Frameworks: Designed and deployed reusable, language-independent data-driven automation frameworks across Web, Desktop (SAP MM), and Mobile applications across Telecom, HR, and Energy domains, automating end-to-end System Integration, Regression, Smoke, and UAT suites. • Stakeholder PoC Engineering: Partnered with business stakeholders to translate complex domain specifications into proof-of-concept (PoC) automation harnesses, accelerating client UAT sign-off velocity.	

APPLIED AI SAFETY RESEARCH, MULTIMODAL FORENSICS & PRACTICAL ENTERPRISE IP (IIT JODHPUR)

Practical Intellectual Property bridging mathematical AI research at IIT Jodhpur directly to enterprise production reliability—solving training data leakage, zero-shot model attribution, synthetic media/document injection, and hallucination bounds.	
• 1. ClozeCongruence (v1–v3): Zero-Shot Foundation Model Provenance & Cross-Lingual AI Forensics (CERN Zenodo / TMLR) Enterprise Vulnerability Addressed (Model Attribution & Global Supply-Chain Governance): Audits third-party LLM gateways against silent model downgrades, shadow AI routing, and unauthorized model substitution across 10 languages and 6 writing scripts (Latin, Cyrillic, CJK, Devanagari, Arabic) with ISO/IEC 27037 cryptographic certificate provenance. [Achieved 100.00% Top-1 attribution accuracy (Macro-F₁ = 1.0000) across 6 frontier LLMs (GPT-4o, Claude 3.5, Gemini 3.7, Nemotron-3 120B, GLM-5.2) with a +97.23% resonance gap, plus 98.92% 10-language AUROC and 0.00% False Accusation Rate on native prose via burstiness modulation (CV_{burst})] CERN Zenodo DOI: 10.5281/zenodo.22158483 SSRN: 7368119 Interactive Artifacts: HF Space (cloze-congruence-ai-forensics-hub-v1) & ScribeMark Live App • 2. Multi-Scale 4-Pass Sentence Cloze Infilling with Pass-Adaptive Dynamic Gating for Robust AI Text Detection Enterprise Vulnerability Addressed (Data Leakage & RAG Corpus Integrity): Defends enterprise RAG ingestion pipelines against synthetic data poisoning, training data leakage, and hallucinated document contamination while guaranteeing a strict 0.00% false-positive quarantine rate on authentic human technical literature. [Zero-shot black-box framework achieving 99.58% AUROC on RAID (ACL 2024), 99.98% AUROC on HC3, and 0.00% False Positive Rate (FPR) on human literature (d = 4.07, p = 7.49e−4) via Kuhn-Munkres S₃ bipartite matching and continuous sigmoidal dynamic gating] CERN Zenodo DOI: 10.5281/zenodo.22158286 SSRN: 7341078 Reproducibility Benchmark Kit (GitHub) • 3. Latent Resonance: Zero-Shot VAE Inversion & Azimuthal Spectral Forensics for Diffusion Image Attribution (IEEE) Enterprise Vulnerability Addressed (Multimodal GenAI & Deepfake KYC Verification): Detects synthetic diffusion imagery (SDXL, Midjourney, DALL-E 3, FLUX) in enterprise KYC/claims pipelines via deterministic VAE latent inversion (σ=0), 2D-FFT deconvolution harmonics (±64, ±128 px), and physical CMOS sensor PRNU noise correlation (p_{RGB}). [Validated on N = 1,000 benchmark images (500 real camera photos vs. 500 diffusion synthetics): 100.0% Detection AUROC, 0.00% False Accusations (0/500 FPR), Cohen's d = 4.84 (p = 2.88e−165), and +3.88 dB PSNR margin] Interactive Artifact: ScribeMark Vision App HF Benchmark Dataset (N=1,000) GitHub Repository • 4. The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Traceback Enterprise Vulnerability Addressed (Compute-Optimal CI/CD Evaluation Gates): Establishes formal parameter scaling laws across 12B to 120B evaluators, enabling LLMops architects to right-size automated judge models for low-latency CI/CD build gates (+51.16% separation gap, Cohen's d = 5.62, p = 2.87e−15). SSRN Series: 7368141 ResearchGate & Preprints.org Open Scaling Datasets & Logs	

EDUCATION & VERIFIED ACADEMIC CHRONOLOGY

Master of Technology (M.Tech) in Augmented & Virtual Reality (AR & VR) Indian Institute of Technology (IIT) Jodhpur — School of Artificial Intelligence & Data Science (Spatial 3D Reconstruction & GANs)	Jul 2022 – Dec 2025 Jodhpur, India
Bachelor of Technology (B.Tech) in Computer Science & Engineering Greater Noida Institute of Technology (GNIOT — AKTU University) [Strict BGV Verified Graduation: June 2014]	Apr 2010 – Jun 2014 Greater Noida, India
Higher Secondary Certificate (Class XII — CBSE [PCMB]) Kendriya Vidyalaya No. 1, Salt Lake (KV No. 1 SLK)	2010 Kolkata, India

CERTIFICATIONS & CORPORATE HONORS

• Technical Certifications: Accenture & MIT Certified Application Tester (ASDA-MIT) • Postman API Testing Path (v12) • IBM Certified Java Professional • Google Crash Course on Python (Coursera). • Corporate Engineering Accolades: Boeing Pride Award (737 MAX Return-to-Service Validation) & Boeing Star Award (2019) • Cognizant Asset Creation & Pride Awards (Reusable Framework & AI Evaluation Utilities) • CIO India Featured Engineering Innovation (One Click Capture at Accenture LLP).	
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